

ACIDS, BASES AND SALTS

Detailed Chemistry Notes

Exam-oriented study material

These notes cover fundamental concepts, theories, strength, pH, indicators, neutralization, salts, common compounds, reactions, formulas, numericals, and important examination questions.

Topic	Coverage
Acids	Definitions, properties, reactions, strength and examples
Bases & Alkalis	Definitions, properties, reactions and differences
Acid–Base Theories	Arrhenius, Brønsted–Lowry and Lewis
pH & Indicators	pH, pOH, Kw, indicators and calculations
Salts	Types, preparation, hydrolysis and common salts
Applications	Baking soda, washing soda, bleaching powder, POP
Exam Preparation	Definitions, reactions, formulas, numericals and questions

1. Introduction

Acids, bases and salts are three important classes of chemical substances. They are involved in neutralization, industrial processes, biological systems, laboratory reactions and many household products.

- Acids generally increase the concentration of hydrogen ions (H^+) in aqueous solution.
- Bases generally increase the concentration of hydroxide ions (OH^-) in aqueous solution.
- Salts are ionic compounds commonly formed when an acid reacts with a base.
- The acidic or basic nature of a solution can be studied using indicators and the pH scale.

2. Acids

Definition

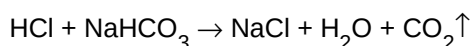
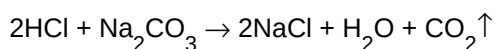
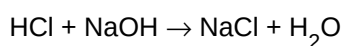
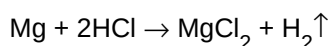
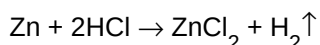
An acid is a substance that donates H^+ ions or increases the hydronium ion concentration in aqueous solution. In Brønsted–Lowry theory, an acid is a proton donor.

Examples: HCl , H_2SO_4 , HNO_3 , CH_3COOH and H_2CO_3 .

Properties of Acids

- Taste: Many common acids have a sour taste; chemicals should never be tasted in the laboratory.
- Indicators: Acids turn blue litmus red.
- pH: Acidic solutions generally have pH less than 7 at 25°C .
- Conductivity: Aqueous acids conduct electricity because they contain ions.
- Reaction with metals: Many acids react with active metals to release hydrogen gas.
- Reaction with carbonates: Acids react with carbonates and bicarbonates to produce carbon dioxide.
- Neutralization: Acids react with bases to form salt and water.

Important Acid Reactions



3. Bases and Alkalis

Definition of a Base

A base is a substance that can accept a proton according to Brønsted–Lowry theory. Many bases produce OH^- ions in aqueous solution.

Examples: NaOH , KOH , $\text{Ca}(\text{OH})_2$, NH_3 and CuO .

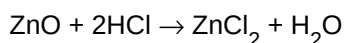
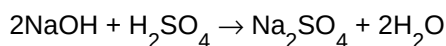
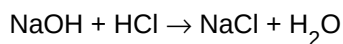
Definition of an Alkali

An alkali is a base that is soluble in water. Examples include NaOH , KOH and aqueous NH_3 .

Properties of Bases

- Soluble bases produce OH^- ions in water.

- Bases turn red litmus blue.
- Basic solutions generally have pH greater than 7 at 25°C.
- They may feel slippery, but chemicals should not be touched deliberately.
- They react with acids during neutralization.
- Some bases react with certain metals such as aluminium or zinc to form complex compounds and release hydrogen.



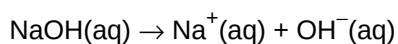
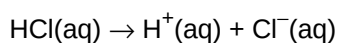
Acid vs Base

Property	Acid	Base
Brønsted definition	Proton donor	Proton acceptor
Litmus	Blue → red	Red → blue
pH at 25°C	< 7	> 7
Typical ion in water	H^+	OH^-
Taste	Often sour	Often bitter; never taste chemicals
Neutralization	Reacts with base	Reacts with acid

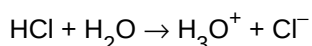
4. Theories of Acids and Bases

4.1 Arrhenius Theory

According to Arrhenius, an acid produces H^+ ions in aqueous solution, while a base produces OH^- ions in aqueous solution.



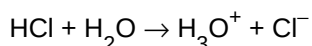
A more accurate representation in water is that free H^+ is associated with water molecules:



Limitation: Arrhenius theory is mainly applicable to aqueous solutions and does not adequately describe acid–base behavior in non-aqueous media.

4.2 Brønsted–Lowry Theory

A Brønsted–Lowry acid is a proton donor, and a Brønsted–Lowry base is a proton acceptor.



Here HCl donates a proton and acts as the acid, while H_2O accepts the proton and acts as the base.

Conjugate Acid–Base Pairs

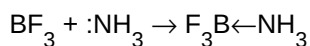
When an acid donates a proton, it forms its conjugate base. When a base accepts a proton, it forms its conjugate acid.



$\text{NH}_3/\text{NH}_4^+$ and $\text{H}_2\text{O}/\text{OH}^-$ are conjugate acid–base pairs.

4.3 Lewis Theory

A Lewis acid accepts an electron pair, whereas a Lewis base donates an electron pair.



BF_3 accepts an electron pair and NH_3 donates an electron pair.

Theory	Acid	Base
Arrhenius	Produces H^+ in water	Produces OH^- in water
Brønsted–Lowry	Proton donor	Proton acceptor
Lewis	Electron-pair acceptor	Electron-pair donor

5. Strong and Weak Acids and Bases

Strong Acids

Strong acids ionize extensively in water. Common examples are HCl , HBr , HI , HNO_3 and H_2SO_4 (for its first ionization).

Weak Acids

Weak acids ionize only partially in water. Examples include CH_3COOH , HF and H_2CO_3 .



Strong and Weak Bases

Strong bases such as NaOH and KOH dissociate extensively in water. Weak bases such as NH_3 react with water only partially.



Strength vs Concentration

Strength describes the extent to which an acid or base ionizes. **Concentration** describes how much solute is present per unit volume. A dilute strong acid can have a lower concentration than a concentrated weak acid; these are different concepts.

6. Acid Dissociation Constant (K_a) and Base Dissociation Constant (K_b)

For a weak acid HA :



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

Larger K_a generally indicates a stronger acid. The corresponding $\text{p}K_a$ is:

$$\text{p}K_a = -\log K_a$$

For a weak base B :



$$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$$

$$\text{p}K_b = -\log K_b$$

Smaller $\text{p}K_a$ means a stronger acid; smaller $\text{p}K_b$ means a stronger base.

7. pH, pOH and Ionic Product of Water

The pH scale is used to express the acidity of a solution. At 25°C, a neutral aqueous solution has pH 7.

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$\text{At } 25^\circ\text{C: } K_w = 1.0 \times 10^{-14}$$

$$\text{At } 25^\circ\text{C: } \text{pH} + \text{pOH} = 14$$

pH	Nature	Typical interpretation
< 7	Acidic	Higher H^+ concentration
= 7	Neutral	H^+ and OH^- balanced
> 7	Basic	Higher OH^- concentration

Solved Numerical 1: Calculate pH

If $[\text{H}^+] = 1 \times 10^{-3} \text{ mol L}^{-1}$:

$$\text{pH} = -\log(1 \times 10^{-3}) = 3$$

Therefore, the solution is acidic.

Solved Numerical 2: Calculate pOH and pH

If $[\text{OH}^-] = 1 \times 10^{-5} \text{ mol L}^{-1}$:

$$\text{pOH} = -\log(1 \times 10^{-5}) = 5$$

$$\text{pH} = 14 - 5 = 9$$

Therefore, the solution is basic at 25°C.

8. Indicators

Indicators are substances that undergo a visible color change over a particular pH range and are used to identify acidic or basic conditions.

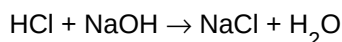
Indicator	Acidic medium	Basic medium
Blue litmus	Red	Blue
Red litmus	Red	Blue
Phenolphthalein	Colorless	Pink
Methyl orange	Red	Yellow
Universal indicator	Red/orange/yellow range	Blue/purple range

Natural Indicators

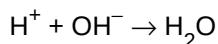
- Litmus is obtained from lichens.
- Turmeric can show a color change in basic conditions.
- Red cabbage extract contains pigments that respond to pH.

9. Neutralization

Neutralization is the reaction between an acid and a base that produces salt and water.



For a strong acid and strong base, the net ionic equation is:



Neutralization is important in antacids, agriculture, wastewater treatment and industrial chemical processing.

10. Salts

Definition

A salt is an ionic compound formed when the replaceable hydrogen ion(s) of an acid are partially or completely replaced by metal ions or another suitable cation. Salts are commonly produced by acid–base reactions.

Types of Salts

Type	Description	Example
Normal salt	Complete replacement of ionizable H of an acid	NaCl , Na_2SO_4
Acid salt	Partial neutralization of a polybasic acid	NaHCO_3 , NaHSO_4
Basic salt	Contains basic hydroxide component	Mg(OH)Cl
Double salt	Crystallizes with two simple salts in fixed ratio	$\text{KAl(SO}_4)_2 \cdot 12\text{H}_2\text{O}$

Hydrated and Anhydrous Salts

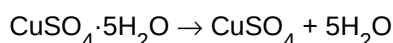
A hydrated salt contains a definite number of water molecules in its crystal structure. An anhydrous salt does not contain this water of crystallization.

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ — hydrated copper(II) sulfate

CuSO_4 — anhydrous copper(II) sulfate

Water of Crystallization

Water of crystallization is the fixed number of water molecules chemically associated with one formula unit of a crystalline salt.

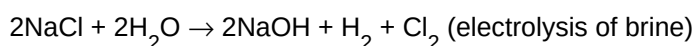


11. Important Common Salts and Compounds

11.1 Common Salt — Sodium Chloride

Formula: NaCl

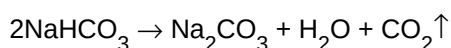
- Used as a food preservative and seasoning.
- Raw material for manufacture of NaOH , H_2 , Cl_2 and other chemicals.



11.2 Baking Soda — Sodium Hydrogen Carbonate

Formula: NaHCO_3

- Used in baking powders and as a mild antacid.



The carbon dioxide released helps dough rise.

11.3 Washing Soda — Sodium Carbonate Decahydrate

Formula: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

- Used as a cleaning agent.
- Used for softening hard water.
- Used in glass, soap and paper industries.

11.4 Bleaching Powder

Common formula: CaOCl_2

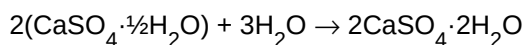
- Used for bleaching textiles and paper.
- Used for disinfection of drinking water.

Bleaching powder is commonly represented as CaOCl_2 ; its exact composition can be more complex depending on preparation and specification.

11.5 Gypsum and Plaster of Paris

Gypsum: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Plaster of Paris: $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$



- POP is used for casts, moulds, decorative work and certain construction applications.

11.6 Copper(II) Sulfate Pentahydrate

Formula: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

- It is commonly called blue vitriol.
- It is used in laboratory demonstrations, agriculture and chemical processes.

12. Salt Hydrolysis

Salt hydrolysis is the reaction of salt ions with water, producing an acidic or basic solution. The pH of a salt solution depends on the strengths of the parent acid and base.

Parent acid	Parent base	Typical salt solution
Strong	Strong	Approximately neutral
Strong	Weak	Acidic
Weak	Strong	Basic
Weak	Weak	Depends on relative K_{a} and K_{b}



Sodium acetate is an example of a salt of a weak acid and strong base; its solution is basic because acetate ions undergo hydrolysis.

13. Buffer Solutions

A buffer solution resists significant changes in pH when small amounts of acid or base are added.

An acidic buffer commonly contains a weak acid and its conjugate base, often supplied by a salt.

Henderson–Hasselbalch: $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$

A basic buffer can be described using:

$$\text{pOH} = \text{pK}_b + \log\left(\frac{[\text{BH}^+]}{[\text{B}]}\right)$$

- Buffers are important in biological systems, analytical chemistry, pharmaceuticals and industrial processes.

14. Titration

Titration is a quantitative analytical technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration.

Equivalence Point vs Endpoint

- **Equivalence point:** The point at which chemically equivalent amounts of reactants have reacted.
- **Endpoint:** The experimentally observed point at which an indicator changes color.

A suitable indicator is selected so that its color-change range lies close to the equivalence point.

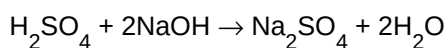
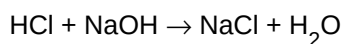
15. Everyday and Industrial Applications

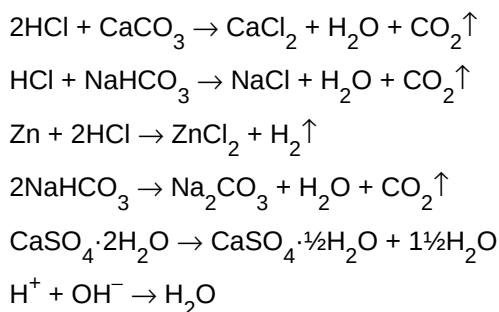
- Antacids neutralize excess stomach acid.
- Agricultural lime is used to reduce excessive soil acidity.
- Toothpaste is generally formulated to help neutralize acids in the mouth.
- Water treatment uses acid–base chemistry to control pH.
- Acid–base reactions are used extensively in chemical manufacturing.
- Salts are used in food preservation, cleaning, construction, medicine, agriculture and industry.

16. Important Formula Sheet

Quantity	Formula
pH	$\text{pH} = -\log[\text{H}^+]$
pOH	$\text{pOH} = -\log[\text{OH}^-]$
Ionic product of water	$K_w = [\text{H}^+][\text{OH}^-]$
At 25°C	$K_w = 1.0 \times 10^{-14}$
At 25°C	$\text{pH} + \text{pOH} = 14$
Acid dissociation	$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$
Base dissociation	$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$
pK _a	$\text{pK}_a = -\log K_a$
pK _b	$\text{pK}_b = -\log K_b$
Conjugate pair relation	$K_a \times K_b = K_w$ (for conjugate pair)

17. Important Reactions for Exams





18. Important Definitions

- **Acid:** A proton donor in Brønsted–Lowry theory.
- **Base:** A proton acceptor in Brønsted–Lowry theory.
- **Alkali:** A water-soluble base.
- **Neutralization:** Reaction of an acid with a base to form salt and water.
- **Salt:** An ionic compound commonly produced through acid–base reactions.
- **Indicator:** A substance that changes color over a characteristic pH range.
- **pH:** Negative logarithm of hydrogen ion concentration in elementary concentration-based calculations.
- **Buffer:** A solution that resists significant pH change when small quantities of acid or base are added.
- **Hydrolysis:** Reaction of salt ions with water that can affect the pH of the solution.

19. Important Exam Questions

Very Short Answer

- Define an acid and a base according to Brønsted–Lowry theory.
- What is an alkali?
- What is neutralization?
- Define pH.
- What is water of crystallization?
- Give two examples of strong acids.
- Give two examples of weak acids.
- What is a conjugate acid–base pair?
- What is a Lewis acid?
- Write the formula of baking soda and washing soda.

Short Answer

- Explain Arrhenius theory and state one limitation.
- Compare Arrhenius, Brønsted–Lowry and Lewis theories.
- Differentiate between strong and weak acids.
- Differentiate between strength and concentration.
- Explain the action of indicators.
- Explain the types of salts with examples.

- Explain the preparation and uses of Plaster of Paris.
- Explain salt hydrolysis with an example.
- Explain the importance of buffers.

Numerical Practice

- Calculate the pH of a solution having $[H^+] = 1 \times 10^{-4}$ M.
- Calculate the pOH and pH of a solution having $[OH^-] = 1 \times 10^{-3}$ M at 25°C.
- If pH = 5, calculate $[H^+]$.
- If pOH = 4 at 25°C, calculate pH.
- For a weak acid with $K_a = 1 \times 10^{-5}$, calculate pK_a .

20. One-Page Quick Revision

Concept	Remember
Acid	Proton donor; acidic solution has pH < 7 at 25°C
Base	Proton acceptor; basic solution has pH > 7 at 25°C
Arrhenius acid	Produces H^+ in water
Arrhenius base	Produces OH^- in water
Lewis acid	Electron-pair acceptor
Lewis base	Electron-pair donor
Neutralization	Acid + Base $\rightarrow$ Salt + Water
pH	$-\log[H^+]$
pOH	$-\log[OH^-]$
At 25°C	pH + pOH = 14
K _w at 25°C	1×10^{-14}
Baking soda	$NaHCO_3$
Washing soda	$Na_2CO_3 \cdot 10H_2O$
Bleaching powder	$CaOCl_2$ (common representation)
Gypsum	$CaSO_4 \cdot 2H_2O$
POP	$CaSO_4 \cdot \frac{1}{2}H_2O$
Blue vitriol	$CuSO_4 \cdot 5H_2O$

21. Final Exam Tips

- Memorize the definitions of acid and base under all three major theories.
- Practice pH, pOH and K_w numericals carefully.
- Learn the formulas, preparation/reactions and uses of common salts.
- Do not confuse acid strength with acid concentration.
- Know the difference between equivalence point and endpoint.

- Balance all chemical equations before using them in answers.
- For theory questions, write definition → explanation → equation/example → application where appropriate.